Corporate Finance

Topic 2 – B

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Taxes

- Consider two firms with the same assets but a different capital structure (mix of debt and equity)

	Levered	Unlevered
EBITDA	2150	2150
Depreciation	300	300
EBIT	1850	1850
Interest expense	350	0
Pre-tax income	1500	1850
Taxes (35%)	**525**	**647.5**
Net income	975	1202.5

- Because interest expenses are **tax deductible**, the levered firm pays less taxes than the unlevered firm

Taxes

- The difference 647.5 – 525 $=$ 122.5 is referred to as the **interest tax shield (ITS)**
- Note that $122.5 = 350 \times 35\% =$ interest expenses $\times \tau$, where τ denotes the corporate taxes
- As interest expenses are equal to the amount of debt, D, times the interest rate, r_D,

$$ITS = \tau \times D \times r_D$$

Taxes

- Does our measure of free cash flow (to the firm) take this tax advantage into account?

	Levered	Unlevered
EBIT(1-35%)	1202.5	1202.5
Depreciation	300	300
FCF	**1502.5**	**1502.5**

- Our definition of FCFs **does not take into account tax benefits of debt**
- They are referred to as **unlevered FCF** because it is as if the firm had no leverage
- We will account for the tax benefits of debt by adjusting the WACC or computing their expected discounted value directly

Taxes

- With taxes, the capital structure affects cash flows accruing to debt and equity holders
- The issue is not tax per se, but differential tax treatment of interest and dividends
- In most countries, debt has a tax advantage at the corporate level because:
  - Interest payments are tax-deductible
  - Dividends and retained earnings are taxed

Taxes

- Consider a firm with a perpetual growth rate g that generates X at $t = 1$ and has a constant level of perpetual safe debt D

- In each period, if the (net) risk-free rate is r_f, the after-tax cash flow is

$$\underbrace{(1-\tau)(X - r_f D)}_{\substack{\text{after-tax cash flow} \\ \text{to equityholders}}} + \underbrace{r_f D}_{\substack{\text{cash flow} \\ \text{to debtholders}}}$$

  which can be rewritten as

$$\underbrace{(1-\tau)X}_{\substack{\text{after-tax cash flow} \\ \text{all-equity, unlevered, firm}}} + \underbrace{\tau r_f D}_{\substack{\text{interest tax} \\ \text{shield (ITS)}}}$$

- In each period the firm can save on taxes the interest tax shield

Taxes

- Thus, relative to an identical unlevered firm, the levered firm can pay out $\tau r_f D$ more to its investors in each period

- Assuming perpetuity

$$V_{ITS} = \frac{\tau r_f D}{r_f} = \tau D$$

and thus the value of the levered firm is

$$V_L = \underbrace{\frac{(1-\tau)X}{r_U - g}}_{V_U} + \underbrace{\tau D}_{V_{ITS}}$$

V_U: value of the hypothetical unlevered firm

where r_U is the rate of return on the unlevered firm's assets (assets of the firm excluding the present value of tax benefits)

Taxes

- **(MM I with corporate taxes)** The value of a levered firm equals that of an unlevered firm plus the present value of the interest tax shield

$$V_L = V_U + V_{ITS}$$

- If investors cannot get an equivalent tax break on homemade leverage, they are willing to pay a premium for levered firms
- All the tax benefits from leverage accrue to the shareholders

Taxes

- Balance sheet with taxes

V_U: market value of the unlevered firm's assets r_U, β_U	Market value of debt D r_D, β_D
V_{ITS} r_{ITS}, β_{ITS}	Market value of equity E r_E, β_E

$$V_L = \underbrace{V_U + V_{ITS}}_{\substack{\text{Assets of} \\ \text{the levered firm}}} = D + E$$

Taxes

- From the balance sheet above, we have the identity

$$\frac{D}{D+E}r_D + \frac{E}{D+E}r_E = \frac{V_U}{V_U+V_{ITS}}r_U + \frac{V_{ITS}}{V_U+V_{ITS}}r_{ITS}$$

- Similarly,

$$\frac{D}{D+E}\beta_D + \frac{E}{D+E}\beta_E = \frac{V_U}{V_U+V_{ITS}}\beta_U + \frac{V_{ITS}}{V_U+V_{ITS}}\beta_{ITS}$$

Taxes

- How do we estimate r_{ITS}?
- $r_{ITS} = r_D$ is the firm maintains a **constant market value debt amount** in perpetuity. In this case,

$$V_{ITS} = \frac{\tau D r_D}{r_D} = \tau D$$

and

$$\frac{D}{D+E} r_D + \frac{E}{D+E} r_E = \frac{V_U}{V_U + \tau D} r_U + \frac{\tau D}{V_U + \tau D} r_D$$

$$\frac{D}{D+E} \beta_D + \frac{E}{D+E} \beta_E = \frac{V_U}{V_U + \tau D} \beta_U + \frac{\tau D}{V_U + \tau D} \beta_D$$

Taxes

- If the firm **continuously adjust** its capital structure to maintain a **constant market debt-to-equity ratio**(target ratio), debt rises with assets and so do tax shields
- Therefore, $\beta_{ITS} = \beta_U$, $r_{ITS} = r_U$ and

$$\frac{D}{D+E} r_D + \frac{E}{D+E} r_E = \frac{V_U}{V_U + V_{ITS}} r_U + \frac{V_{ITS}}{V_U + V_{ITS}} r_U = r_U$$

$$\frac{D}{D+E} \beta_D + \frac{E}{D+E} \beta_E = \frac{V_U}{V_U + V_{ITS}} \beta_U + \frac{V_{ITS}}{V_U + V_{ITS}} \beta_U = \beta_U$$

- This is the usual assumption in valuation

Taxes

- Three closed-form formulas for a firm whose unlevered free cash flows grow at a rate g in perpetuity (DeMarzo, 2005; Jagannathan et al., 2017; Stanton and Seasholes, 2005):
  - Constant debt level:
$$V_{ITS} = \tau D$$
  - Constant debt-to-equity ratio (annually adjusted):
$$V_{ITS} = \left(\frac{\tau r_D D}{r_u - g}\right)\left(\frac{1 + r_U}{1 + r_g}\right)$$
  - Constant debt-to-equity ratio (continuously adjusted):
$$V_{ITS} = \frac{\tau r_D D}{r_u - g}$$

Example

- Example: an unlevered firm with a value of 600 and 1000 shares (0.6 per share) plans to (safely) borrow (perpetuity) debt to immediately repurchase 500 worth of shares
- Assume $\tau = 40\%$
- We have seen that the value of the levered firm is $600 + 40\% \times 500 = 800$
- What would be the new share price?
- The value of equity would be 800 – 500 $=$ 300, and this value must be equal to the new share price P times the number or outstanding shares, 1000 – N, where N is the number of shares repurchased

Example

- Under competition and perfect information, share repurchases are zero NPV transactions, and investors must be indifferent between selling or keeping their shares

$$\underbrace{500}_{\text{amount raised}} = P \times N$$

- As $PN = 500$ and $300 = P(1000 - N)$, $P = 0.8$
- Each shareholder will become richer by $0.8 - 0.6 = 0.2$
- As there are initially 1000 shares, the total wealth gain to shareholders is $0.2 \times 1000 = 200$, which is exactly V_{ITS}

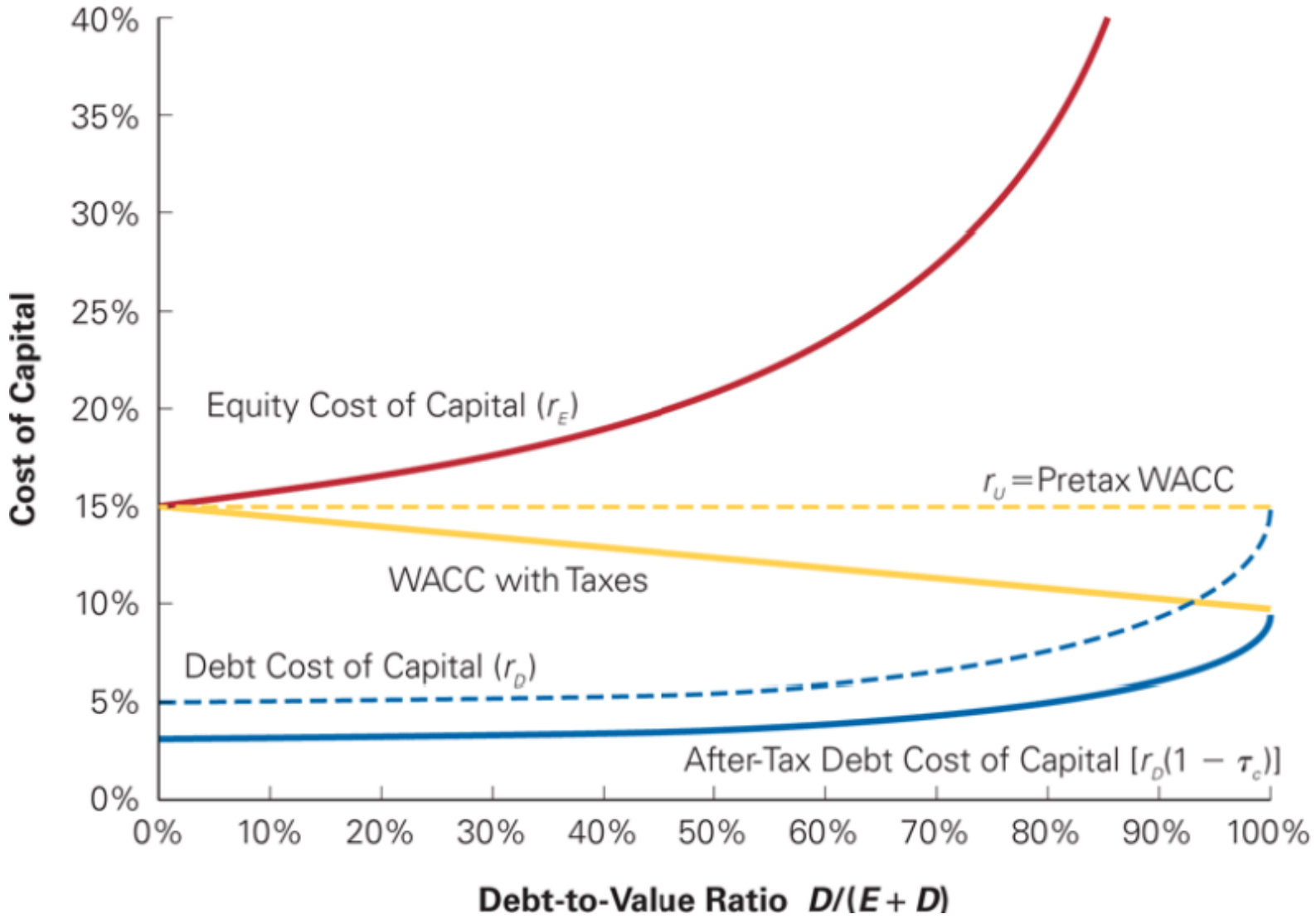
Taxes

- With tax-deductible interest, the effective return on debt if $r_D(1-\tau)$
- If the firm continuously adjust its capital structure to keep a constant (target) debt-to-equity ratio, the WACC with corporate taxes (**the after-tax WACC**) is

$$r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D(1-\tau)$$

- If we discount the **unlevered free cash flows** with the **after-tax WACC**, we obtain the correct value of the levered firm

Taxes



Source: Berk and DeMarzo (2011)

WACC: three cases

- Three closed-form formulas for a firm whose unlevered free cash flows grow at a rate g in perpetuity (DeMarzo, 2005; Jagannathan et al., 2017; Stanton and Seasholes, 2005):
  - Constant debt level:

$$r_{WACC} = \left(1 - \tau \frac{D}{D+E}\right) r_U$$

  - Constant debt-to-equity ratio (annually adjusted):

$$r_{WACC} = r_U - \tau r_D \left(\frac{D}{D+E}\right)\left(\frac{1+r_U}{1+r_g}\right)$$

  - Constant debt-to-equity ratio (continuously adjusted):

$$r_{WACC} = r_U - \frac{\tau r_D D}{D+E} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1-\tau)$$

Valuation with taxes

- Two methods that take into account the tax advantage of debt: the **adjusted present value (APV)** method and the **weighted average cost of capital (WACC)** method
- The APV method is theoretically more flexible, but less commonly used
- The WACC method is more commonly used, but it requires strong assumptions

The adjusted present value (APV) method

- Steps:
  - Compute the project's unlevered after-tax free cash flows
  - Estimate the unlevered asset β using the comparables approach
  - Find the value of the unlevered firm: the present value of these cash flows using the estimated unlevered asset β
  - Find the V_{ITS} and add to the value of the unlevered firm

The adjusted present value (APV) method

- V_{ITS}:
  - If the project is partially financed with debt, the V_{ITS} is the present value of tax savings on current and future **incremental** interest payments
  - If the increase in debt is not permanent, calculate the V_{ITS} as an annuity rather than a perpetuity
  - Use the appropriate r_{ITS}
  - Use the marginal tax rate, that is, the tax rate of the firm including the project

The weighted average cost of capital (WACC) method

- Steps:
  - Compute the project's unlevered after-tax free cash flows
  - Estimate the unlevered asset β_U using the comparables approach
  - Relever that asset β using the target debt-to-equity ratio to get the expected return on equity
  - Compute the project's debt cost of capital (using debt returns on similarly related projects)
  - Compute the after-tax WACC

The weighted average cost of capital (WACC) method

- If FCF_t denotes the unlevered free cash flow in period t, the value of the levered firm is

$$V_L = \frac{FCF_1}{1 + r_{WACC}} + \frac{FCF_2}{(1 + r_{WACC})^2} + \frac{FCF_3}{(1 + r_{WACC})^3} + \ldots$$

- If leverage changes a lot over time, we should compute r_{WACC} for different periods separately

APV and WACC

	Cash flows	Discount rate
APV	Unlevered free cash flow Interest tax shield	Unlevered (asset) cost of capital, r_U Tax shield cost of capital, r_{ITS}
WACC	Unlevered free cash flow	After-tax WACC, r_{WACC}

Equivalency of the APV and WACC methods when D/E is constant

- Suppose that the unlevered FCF one period from now is equal to FCF_1 and that this cash flow will grow in perpetuity at a growth rate g
- Using the WACC method, we have that

$$V_L = \frac{FCF_1}{r_{WACC} - g}$$

- Using the formula for the after-tax WACC,

$$V_L = \frac{FCF_1}{\frac{E}{V_L} r_E + \frac{D}{V_L} r_D (1 - \tau) - g}$$

Equivalency of the APV and WACC methods when D/E is constant

- When D/E is constant,

$$\frac{E}{V_L} r_E + \frac{D}{V_L} r_D = r_U$$

so

$$V_L = \frac{FCF_1}{r_U - \frac{D}{V_L} r_D \tau - g}$$

- Rearranging

$$V_L (r_U - g) - D r_D \tau = FCF_1$$

Equivalency of the APV and WACC methods when D/E is constant

- Then

$$V_L = \underbrace{\frac{FCF_1}{r_U - g}}_{=V_U} + \underbrace{\frac{Dr_D\tau}{r_U - g}}_{=V_{ITS}}$$

- The APV and WACC methods are equivalent when D/E is constant

Equivalency of the APV and WACC: example

- You are considering investing 60,000 in a project today:
  - Incremental unlevered FCF in period 1 is 7,500 and then it will grow at a 4% rate thereafter
  - Tax rate, τ, is 40%
  - Target leverage ratio of the project is $D/(D+E)$ is 25% (not necessarily the same as the target leverage of the firm)
  - $r_E = 14.33\%$, $r_D = 5\%$ (pre-tax)

Year	0	1	2	3	...
Unlevered FCF	-60,000	7,500	7,800	8,112	...

Equivalency of the APV and WACC: example

- The after-tax WACC is

$$r_{WACC} = 75\%14.33\% + 25\%5\%(1 - 40\%) = 11.5\%$$

- The value of the levered firm using the WACC method is

$$V_L = \frac{FCF_1}{r - g} = \frac{7,500}{11.5\% - 4\%} = 100,000$$

- The NPV of the project is 100,000-60,000=40,000

Equivalency of the APV and WACC: example

- Given that $V_L = 100,000$ and $D/V_L = 25\%$, $D = 25,000$
- The unlevered (asset) cost of capital, r_U, is

$$r_U = 75\% \times 14.33\% + 25\% \times 5\% = 12\%$$

- The value of the unlevered firm is

$$V_U = \frac{FCF_1}{r_U - g} = \frac{7,500}{12\% - 4\%} = 93,750$$

Equivalency of the APV and WACC: example

- As the firm is growing at 4% and the proportion of debt is constant, debt is also growing at 4% and so does ITS
- The present value of ITS is

$$V_{ITS} = \frac{Dr_D\tau}{r_U - g} = \frac{25,000 \times 5\% \times 40\%}{12\% - 4\%} = 6,250$$

- The value of the levered firm using the APV method is then

$$V_L = V_U + V_{ITS} = 93.750 + 6,250 = 100,000$$

- The NPV is 40,000

APV: flexibility when D/E is not constant

- Suppose now a different leverage policy: the firm will borrow 60,000 at $t = 0$, and repay 60,000 in 5 years
- The value of the unlevered firm does not change: $V_U = 93,750$
- What about V_{ITS}?

Year	0	1	2	3	4	5
Net borrowing	60,000					-60,000
Interest (5%)		-3,000	-3,000	-3,000	-3,000	-3,000
ITS (40%)		1,200	1,200	1,200	1,200	1,200
V_{ITS} at 5%	5,195					

- In this case, $V_L = 93,750 + 5,195 = 98,945$

Taxes

- Suppose that an unlevered firm issues debt level D, which will remain fixed in perpetuity, to repurchase equity
- If $\tau = 35\%$, then

$$V_L = V_U + 35\%D \quad \Rightarrow \frac{V_L}{V_U} = 1 + \frac{35\% \times D}{V_U}$$

- For $D = 20\%V_U$, the value of the firm increases by 7%; for $D = 50\%V_U$, the value of the firm increases by 17.5%
- It seems that the optimal capital structure is attained with 100% debt
- Obviously this is not true

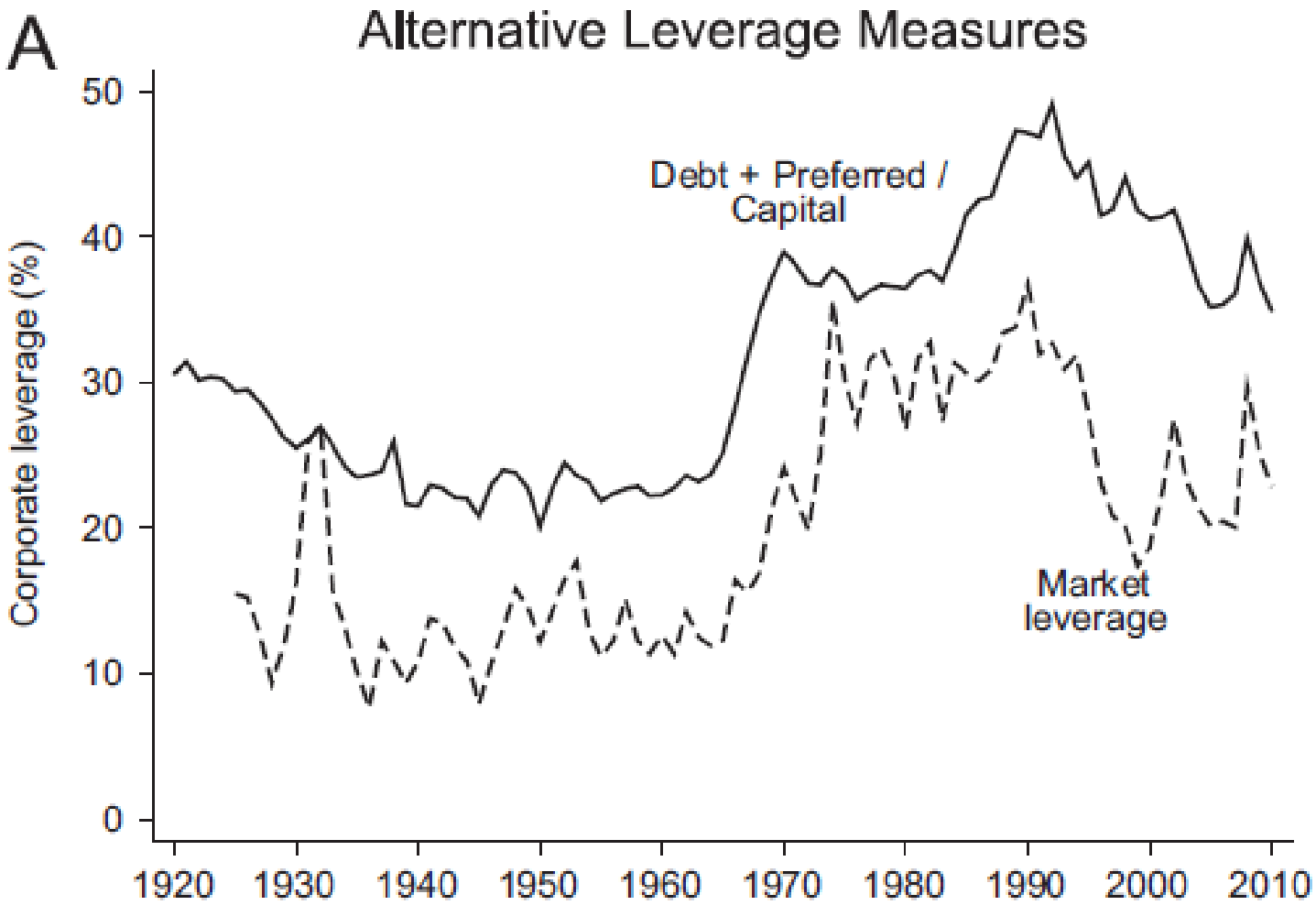
Jagannathan et al. (2017)

Table 5 Weighted average cost of capital (WACC) across industries in 2016

Variable	Durable	Energy	High-tech	Health	Manufacturing	Nondurable	Shops	Telecom	Utilities	Other
Market capitalization	333	1,486	4,782	2,193	2,240	1,511	2,301	1,287	999	2,271
Market capitalization for firms with rating	218	1,246	3,486	1,627	1,767	1,304	1,750	994	856	1,183
% Coverage by market capitalization	65.5%	83.8%	72.9%	74.2%	78.9%	86.3%	76.0%	77.2%	85.7%	52.1%
Number of firms	78	154	569	544	389	135	318	77	101	389
Number of firms with rating	12	26	59	23	77	34	43	10	66	50
% Coverage by number of firms	15.4%	16.9%	10.4%	4.2%	19.8%	25.2%	13.5%	13.0%	65.3%	12.9%
β_S	1.42	1.25	0.98	0.74	0.97	0.53	0.77	0.68	0.53	1.06
T_C (%)	30.71	22.35	20.60	18.35	26.60	27.77	33.84	35.69	32.32	25.37
$E[R_S]$	11.74	10.65	8.87	7.28	8.78	5.92	7.52	6.90	5.93	9.32
$E[R_D]$	4.61	4.33	4.22	4.28	4.29	4.42	4.34	4.55	4.61	4.49
$D/(D+S)$	0.44	0.18	0.12	0.15	0.16	0.18	0.14	0.27	0.41	0.23
$S/(D+S)$	0.56	0.82	0.88	0.85	0.84	0.82	0.86	0.73	0.59	0.77
WACC	8.02	9.05	8.21	6.66	7.78	5.40	6.85	5.86	4.69	7.93

This table reports summary statistics for WACC and its components across the 10 Fama–French industries in 2016. All variables are defined in **Table 2**. We compute the average of the firm-specific measures weighted by the market capitalization of each firm. In aggregating the components to compute WACC, we restrict the sample to only those firms that have a WACC calculated in **Table 2**. The definitions of the Fama–French industries can be found at Kenneth French's website: **http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_10_ind_port.html**. The market capitalization of all Compustat/CRSP US incorporated nonfinancial firms is shown in billions of dollars. Market capitalization for firms with ratings represents the market capitalization of firms with bond yield data available in billions of dollars. Also shown are the total number of firms and the number of firms for which some of the WACC component data are not available (mostly bond yield data).

Graham et al. (2015)



Bankruptcy costs

- More leverage increases the risk of financial distress and bankruptcy
- A bankruptcy process transfers ownership to debtholders
- This transfer of control is not a violation of MM per se
- The violation comes from **bankruptcy costs**:
  - **Direct costs**
  - **Indirect costs**
- Such bankruptcy costs may countervail the tax advantage of debt financing

Direct bankruptcy costs

- Ex post or direct costs are incurred conditional on actual liquidation or reorganization:
  - Legal and administrative costs
  - Time and resources spent by managers and creditors dealing with bankruptcy
  - Fire sales of assets due to asset specificity and illiquid assets

Indirect bankruptcy costs

- Ex ante or indirect costs can take place even before a firm declares bankruptcy
- Decrease in demand, especially in industries that rely on long-term relationships with customers such as airlines (frequent flier programs) and automobiles (warranties, replacement parts, and maintenance)
  - Hortaçsu et al. (2013): *An increase in a manufacturer's credit default swaps significantly decreases the prices of its cars at auction, especially cars with longer expected service lives.*
- Inability to access credit from suppliers and banks (due to credit market imperfections)

Indirect bankruptcy costs

- Loss of current employees; more difficult to attract new ones
  - Risk-averse workers prefer stable jobs and want to avoid the scarring effects of unemployment
  - Brown and Matsa (2016): *Firms attract significantly fewer applications per job opening during periods of corporate distress.*
  - Baghai et al. (2021): *Firms lose workers with the highest cognitive and noncognitive skills as they approach bankruptcy. (...) Consistent with talent dependence being associated with higher labor costs of financial distress, firms that rely more on talent have more conservative capital structures.*
  - Van Doornik et al. (2022): *More generous unemployment insurance (UI) system shifts labor supply from safer to riskier firms and reduces the compensating wage differentials that risky firms need to pay.*

Indirect bankruptcy costs

- Conflict of interest between equityholders and creditors might lead to poor investment decisions:
  - Debt overhang: underinvestment
  - Asset substitution or risk-shifting: excessive risk-taking
  - Cashing out: shareholders might try to withdraw cash from the firm in anticipation of losing it to senior claimants

Trade-off theory

- In the presence of bankruptcy costs, the optimal capital structure trades-off the tax advantage of debt versus the costs of financial distress

- Consider a firm that has at $t = 0$ debt outstanding that requires a total repayment of K at $t = 1$

- At $t = 1$:

  - The firm generates random cash flows X distributed over the interval $[\underline{X}, \bar{X}]$ with a p.d.f $h(X)$ and an associated c.d.f. $H(a) = Pr(X \leq a) = \int_{\underline{X}}^{a} h(x)dx$

  - The full amount K is tax deductible (for simplicity) and the firm pays taxes τ if solvent ($X \geq K$)

  - Bankruptcy ($X < K$) imposes a fixed cost $C < \underline{X}$

Trade-off theory

- All investors are risk-neutral and require an expected return of r_f
- The market value of the unlevered firm is

$$V_U = \frac{1}{1+r_f}\int_{\underline{X}}^{\bar{X}}(1-\tau)Xh(X)dX = \frac{(1-\tau)\mathbb{E}[X]}{1+r_f}$$

- With risk-free debt $K < \underline{X}$, the firm's market value is

$$\begin{aligned} V_L(K) &= \frac{1}{1+r_f}\int_{\underline{X}}^{\bar{X}}[(1-\tau)(X-K)+K]h(X)dX \\ &= \frac{(1-\tau)\mathbb{E}[X-K]}{1+r_f} + \frac{K}{1+r_f} = V_U + \frac{\tau K}{1+r_f} \end{aligned}$$

Trade-off theory

- With risky debt, the market value of debt is

$$E(K) = \frac{1}{1+r_f}\int_K^{\bar{X}} (1-\tau)(X-K)h(X)dX$$

and the market value of debt is

$$\begin{aligned} D(K) &= \frac{1}{1+r_f}\int_{\underline{X}}^{K} (X-C)h(X)dX + \frac{1}{1+r_f}\int_K^{\bar{X}} Kh(X)dX \\ &= \frac{1}{1+r_f}\int_{\underline{X}}^{K} (X-C)h(X)dX + \frac{K[1-H(K)]}{1+r_f} \end{aligned}$$

Trade-off theory

- Adding $E(K)$ and $D(K)$ yields

$$V(K) = V_U + \frac{1}{1+r_f}\left[\tau\left(\int_{\underline{X}}^{K} Xh(x)dx + K[1-H(K)]\right) - CH(K)\right]$$

- An increase in K has two effects
  - It increases the value of the tax shield

$$\tau\left(\int_{\underline{X}}^{K} Xh(x)dx + K[1-H(K)]\right)$$

  - But it increases the expected bankruptcy cost C: $CH(K)$

Trade-off theory

- The optimal debt level K^* is such that the marginal benefit of debt equals the marginal expected bankruptcy costs
- K^* increases with the tax benefits τ
- K^* decreases with bankruptcy costs C
- With bankruptcy costs,

$$V_L = V_U + V_{ITS} - PV(\text{bankruptcy costs})$$

- We can adjust the APV method to consider PV(bankruptcy costs)

Application: the debate about optimal bank capital regulation

- Heated debate about tighter financial regulation following the 2007-2008 financial crisis, including higher capital requirements
- Banks: negative consequences to credit markets and, consequently, economic growth
- Admati et al. (2013): no!
  - Bank equity is not *socially* expensive
  - They use many of the arguments discussed in this lecture
  - They call for larger capital requirements, that is, less leverage in the banking industry

Application: the debate about optimal bank capital regulation

- True or false?
  - Statement 1: “Increased equity requirements will hurt bank shareholders since it would lower the banks’ return on equity (ROE).”
  - Statement 2: “Increased equity requirements increase the funding costs for banks because they must use more equity, which has a higher required return.”
  - Statement 3: “Increased equity requirements increase the funding costs for banks because they reduce the ability of banks to benefit from the tax shield associated with interest payments on debt.”

    ... negative externality because the increase in leverage raises the probability of a bank failure, weakening the financial system and imposing losses on the broader economy.

MM - dividend irrelevance

- The MM logic applies to all aspects of financial policy as all purely financial transactions have a zero NPV
  - Other financing instruments such as convertible bonds, preferred stocks, ...
  - Choice between long-term and short-term debt
  - Risk management
- **(MM dividend irrelevance proposition)** A firm's total market value is independent of its dividend policy

MM - dividend irrelevance

- If the investment policy is fixed, a given change in the payout policy has to be met by a change in the financing policy
- For example, an increase in dividends has to be financed with a new/greater equity or debt issue
- Consider an all-equity firm with risk-neutral investors
- At each time $t = 1, 2, ...$ the firm generates some random cash flow X_t and
  - pays a per-share dividend D_t
  - makes an investment I_t
  - has n_t shares outstanding

MM - dividend irrelevance

- Let δ be the discount factor and P_t the share price at the end of period t (after the cash flow is realized, dividends paid and share number adjusted)
- We have that $P_t = \delta \mathbb{E}[D_{t+1} + P_{t+1}]$
- New shareholders in year t do not receive D_t, so they buy shares at a price P_t. It must be the case that

$$\underbrace{n_{t-1} D_t + I_t}_{\text{use of funds}} = \underbrace{X_t + \overbrace{P_t(n_t - n_{t-1})}^{\substack{\text{amount from} \\ \text{the equity issue}}}}_{\substack{\text{sources of funds} \\ \text{(earned or raised)}}}$$

MM - dividend irrelevance

- The total share value at the end of period t is

$$V_t = n_t P_t = \delta \mathbb{E}[n_t D_{t+1} + n_t P_{t+1}]$$

- Using the equality between uses and sources of funds for the period $t+1$, we get

$$V_t = n_t P_t = \delta \mathbb{E}[X_{t+1} + P_{t+1}(n_{t+1} - n_t) - I_{t+1} + n_t P_{t+1}]$$

$$= \delta \mathbb{E}[X_{t+1} - I_{t+1} + V_{t+1}]$$

- Solving recursively

$$V_t = \mathbb{E}\left[\sum_{i=1}^{+\infty} \delta^i (X_{t+1} - I_{t+1})\right]$$

- The value of the all-equity firm does not depend on the payout policy as long as the investment plan I_t is fixed

- Result extends to debt issuance and risk-aversion

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Appendix: WACC derivation

- As $V_L = D + E = V_U + V_{ITS}$, the expected return on the assets (= assets of the unleverd firm + PV of tax shields) of the levered firm is

$$r_L = \frac{D}{D+E} r_D + \frac{E}{D+E} r_E$$

- If we assume that
  - the firm keeps a constant leverage ratio (continuously adjusted)
  - r_U, the expected return on the unlevered firm, is not time-varying (same for all t)

  we have that, for any period, $r_L = r_U$

- See Stanton and Seasholes (2005) and DeMarzo (2005) for a formal treatment

Appendix: WACC derivation

- The (expected) cash flow available to creditors and shareholders in period $t+1$,

$$\underbrace{C_{L,t+1}}_{\substack{\text{cash flow available to} \\ \text{creditors and shareholders}}} = \underbrace{FCF_{t+1}}_{\substack{\text{unlevered free} \\ \text{cash flow}}} + \underbrace{r_d D_t \tau}_{\substack{\text{interest tax} \\ \text{shield}}}$$

- Because the investment continues after period $t+1$, there is a continuation value $V_{L,t+1}$

- The value of the firm in period t is

$$V_{L,t} = \frac{FCF_{t+1} + r_d D_t \tau + V_{L,t+1}}{1 + r_L}$$

Appendix: WACC derivation

- The same is true for the period $t+1$

$$V_{L,t+1} = \frac{FCF_{t+2} + r_d D_{t+1} \tau + V_{L,t+2}}{1 + r_L}$$

- Plugging this expression for $V_{L,t+1}$ into the expression for $V_{L,t}$, we have that

$$V_{L,t} = \frac{FCF_{t+1} + r_d D_t \tau}{1 + r_L} + \frac{FCF_{t+2} + r_d D_{t+1} \tau + V_{L,t+2}}{(1 + r_L)^2}$$

$$= \frac{C_{L,t+1}}{1 + r_L} + \frac{C_{L,t+2}}{(1 + r_L)^2} + \frac{V_{L,t+2}}{(1 + r_L)^2}$$

Appendix: WACC derivation

- Plugging the expression for $V_{L,t+2}$ into the expression for $V_{L,t}$, we have that

$$V_{L,t} = \frac{C_{L,t+1}}{1+r_L} + \frac{C_{L,t+2}}{(1+r_L)^2} + \frac{C_{L,t+3}}{(1+r_L)^3} + \frac{V_{L,t+3}}{(1+r_L)^3}$$

- Doing these operations N times, we have that

$$V_{L,t} = \frac{C_{L,t+1}}{1+r_L} + ... + \frac{C_{L,t+N}}{(1+r_L)^N} + \frac{V_{L,t+N}}{(1+r_L)^N}$$

- When $N \to \infty$, the term

$$\frac{V_{L,t+N}}{(1+r_L)^N} \to 0$$

Appendix: WACC derivation

- Thus

$$V_{L,t} = \frac{C_{L,t+1}}{1+r_L} + \frac{C_{L,t+2}}{(1+r_L)^2} + \frac{C_{L,t+3}}{(1+r_L)^3} + ...$$

$$= \sum_{i=1}^{\infty} \frac{C_{L,t+i}}{(1+r_L)^i}$$

- This is the standard valuation formula for an infinitely-lived asset (going concern)
- In practice, it is not used: we need an estimate of the interest tax shield for all periods
- The WACC method simplifies our life

Appendix: WACC derivation

- Recall that

$$V_{L,t} = \frac{FCF_{t+1} + r_d D_t \tau + V_{L,t+1}}{1 + r_L}$$

and

$$r_L = \frac{D_t}{D_t + E_t} r_D + \frac{E_t}{D_t + E_t} r_E$$

- Then

$$V_{L,t}(1 + r_L) - r_d D_t \tau = FCF_{t+1} + V_{L,t+1}$$

$$(\Rightarrow) \quad V_{L,t}(1 + r_L) - V_{L,t} r_d \frac{D_t}{V_{L,t}} D_t \tau = FCF_{t+1} + V_{L,t+1}$$

$$(\Rightarrow) \quad V_{L,t}\left(1 + r_L - r_d \frac{D_t}{V_{L,t}} \tau\right) = FCF_{t+1} + V_{L,t+1}$$

Appendix: WACC derivation

$$(\Rightarrow) \quad V_{L,t} = \frac{FCF_{t+1} + V_{L,t+1}}{1 + r_L - r_d \frac{D_t}{V_{L,t}} \tau}$$

- But note that

$$\begin{aligned}
1 + r_L - r_d \frac{D_t}{V_{L,t}} \tau &= 1 + \frac{D_t}{D_t + E_t} r_D + \frac{E_t}{D_t + E_t} r_E - r_d \frac{D_t}{V_{L,t}} \tau \\
&= 1 + \frac{D_t}{D_t + E_t} r_D + \frac{E_t}{D_t + E_t} r_E - r_d \frac{D_t}{D_t + E_t} \tau \\
&= 1 + \frac{D_t}{D_t + E_t} r_D (1 - \tau) + \frac{E_t}{D_t + E_t} r_E \\
&= 1 + r_{WACC}
\end{aligned}$$

Appendix: WACC derivation

- Then

$$V_{L,t} = \frac{FCF_{t+1} + V_{L,t+1}}{1 + r_{WACC}}$$

- Doing the same interactions as before, we can show that

$$V_{L,t} = \frac{FCF_{L,t+1}}{1 + r_{WACC}} + \frac{FCF_{L,t+2}}{(1 + r_{WACC})^2} + \frac{FCF_{L,t+3}}{(1 + r_{WACC})^3} + \ldots$$

$$= \sum_{i=1}^{\infty} \frac{FCF_{L,t+i}}{(1 + r_{WACC})^i}$$

- We only need to project the unlevered free cash flows if we know the r_{WACC}

Appendix: Comment on r_D

- If debt is risky, is the interest tax shield equal to $r_D D_t \tau$?
- Suppose that a firm issued a bond in the past at par with an interest rate (=coupon payment) of 5% and a face value of 1000
- The coupon payment is $5\% \times 1000 = 50$
- The bond matures in the next period
- The expected return on the debt, r_D is

$$D_t = \frac{\mathbb{E}[C_{t+1} + D_{t+1}]}{1 + r_D} \iff r_D = \frac{C_{t+1} + D_{t+1} - D_t}{D_t}$$

where $\mathbb{E}[C_{t+1}] = 50$ and $\mathbb{E}[D_{t+1}] = 1000$ if the bond is risk-free

Appendix: Comment on r_D

- Suppose the debt is risky
- For instance, $\mathbb{E}[C_{t+1} + D_{t+1}] = 700 < 1050$, and the market value of debt is 525
- This means that $r_D = 33.33\%$
- Is the expected interest payment equal to $525 \times 33\% = 175$?
- No. The return that determines the interest expense is

$$\bar{r}_D = \frac{\mathbb{E}[C_{t+1}]}{D_t}$$

and the after-tax cost of debt is $r_D - \tau \bar{r}_D$